

Resilience of Networks in the B5G Era

Toward dynamic assessment of availability with Digital Twins

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Orange Research, Châtillon, France

September 14th, 2026



Acknowledging the work of a whole team: current and past members

Current members from Orange team:

- Guillaume FRAYSSE, PhD
- Bertrand DECOCQ, PhD
- Benoit LEMOINE
- Firas MAALEJ, intern, soon to be PhD student
- Christian TANGUY, PhD, HDR
- Ikram TEMERICHT, PhD student

Past members:

- Rui LI, PhD
- Khaled SAYAD, PhD
- Manelle NOUAR



Acknowledging the work of a whole team: collaborations

Risk and Resilience of Complex Systems (RRSC)
chaire at CentraleSupélec:

- Anne BARROS
- Zhiguo ZENG
- Yiping FANG

Collaborative project:

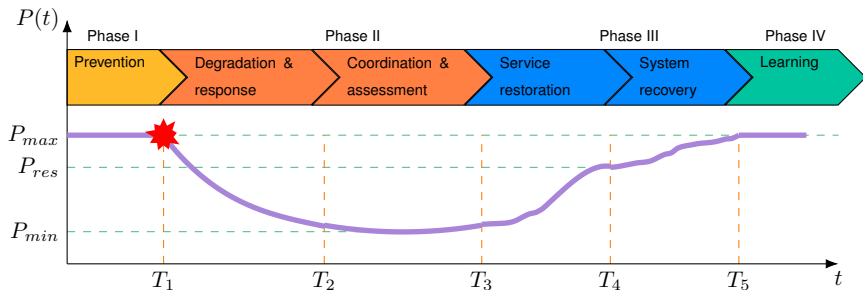
- SNS-JU Programmable Security Plane for 6G Ecosystems (MARE)



Definition

Resilience of networks (from ResiliNets project <https://resilinetts.org>)

Definition: *Resilience* is the ability of the network to face various incidents, maintain an acceptable level of service and return to normal operation.



from Nait Belaid et al. [1]

How do we measure resilience?

Availability

A

Is the service operational ?

Uptime / downtime

Reliability

MTBF

How often does it fail?

MTBF / MTTF

Recoverability

MTTR

How fast can it be restored?

MTTR / TTR

Service continuity

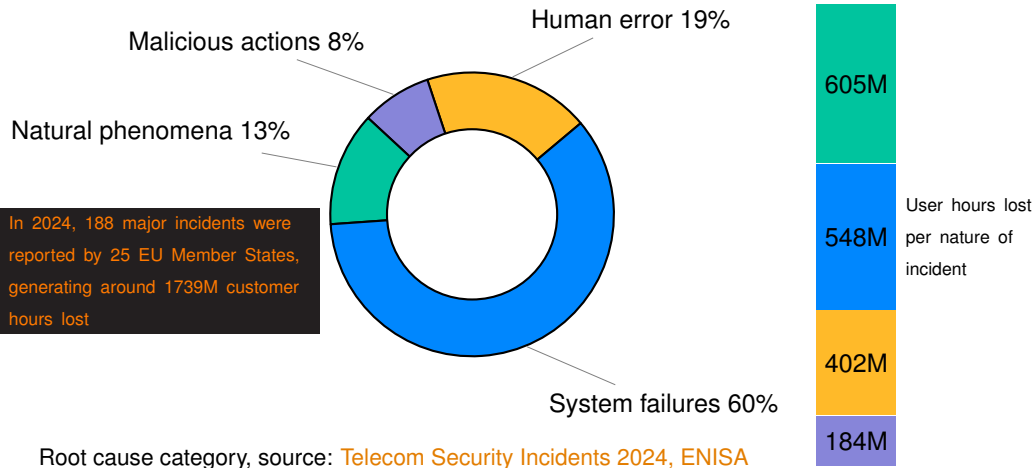
QoS

How much does service degrade?

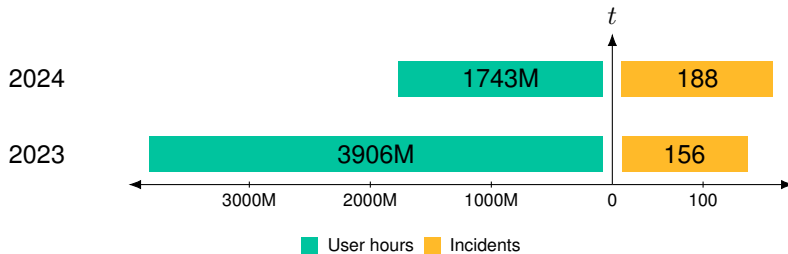
Latency / loss / throughput**Resilience**
assessment

$$A = \frac{\text{Uptime}}{\text{Observation time}}$$

Root Causes of incidents and their impact (2024)



YoY evolution of incidents

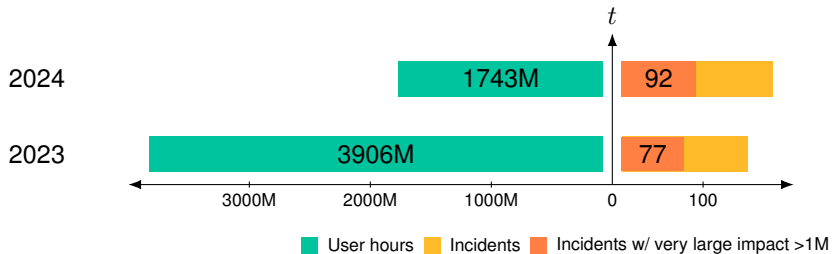


Number of incidents submitted by countries and user hours lost (2023–2024)

source: [Telecom Security Incidents 2024, ENISA](#)



YoY evolution of incidents



Number of incidents submitted by countries and user hours lost (2023–2024)

source: [Telecom Security Incidents 2024, ENISA](#)



HLR (Home Location Register) incident in France July 2012

HLR Headaches

7:00 AM Are subscriber database issues going to bedevil mobile operators?

July 25, 2012

4 Min Read



7:00 AM -- Are subscriber database issues going to become the focal point for mobile operator headaches?

Recent evidence suggests that updates, upgrades and new deployments of HLRs (home location registers), HSSs (home subscriber servers -- like a HLR on steroids) and unifying subscriber databases can cause major service availability problems that cost money, affect credibility and potentially be the catalyst for churn.

- Telefónica UK Ltd. , better known as O2, July 2012: A major service outage affected more than 7 million customers. Neither the operator nor its principal supplier Ericsson AB (Nasdaq: ERIC) have provided any detailed insight -- both have said they have been jointly looking into the cause of the disruption -- but the problems appear to have lay somewhere within its subscriber database infrastructure, either with its HLR servers or supporting shared database (one that is then used by multiple platforms such as the HLR, policy control server, etc). The operator has instigated a charm offensive to appease its customers but isn't providing any further details about the outage, other than to say that it experienced an "unprecedented software fault on the register that works to connect mobile devices to our network. The network is now stable and we are already implementing changes to mitigate the risk of this happening again." O2's services are going to be under specific scrutiny in the coming weeks as it is the official mobile services provider within the London Olympic Village. (See Outage Strikes O2 UK, Now What, O2? and O2's Payback.)

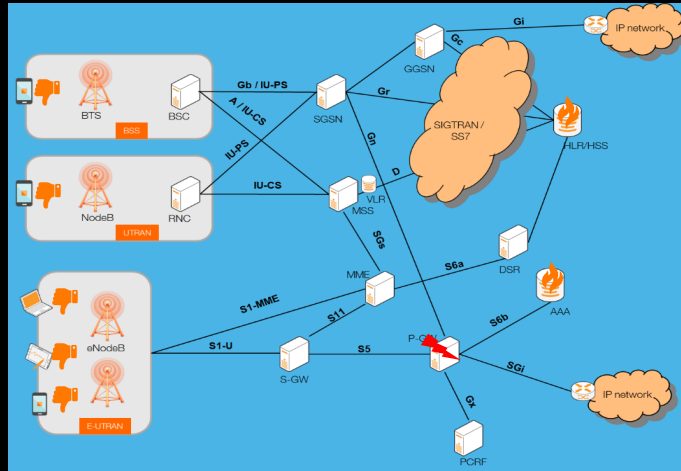
- Orange France , July 2012: A major Friday evening service outage that affected about 26 million customers followed a software upgrade to Alcatel-Lucent (NYSE: ALU) HLR software carried out by staff from the vendor and the carrier. The upgrade introduced data inconsistencies that led to a rising tide of inconsistent messages being shared by Orange France's subscriber systems, which eventually became overloaded and unable to process new requests. The operator provided details of why and how the outage happened in testimony to the country's National Assembly: France takes these issues seriously.

- Telenor Group (Nasdaq: TELN), June 2011: An upgrade to a mobile broadband-related server resulted in the "most extensive breakdown that Telenor has experienced since the mobile network was established in 1993." Signaling traffic between various servers escalated and then "created disturbances" in other network elements, affecting the availability of voice and text message services to 3 million users. (See Telenor Explains Mobile Outage.)

- Verizon Wireless , April 2011: The U.S. operator suffered problems with its 4G/LTE service that were linked to a Nokia Networks HSS. Verizon has subsequently suffered a number of interruptions to its 4G services, but the problems appear to have been caused by a different network issue each time. (See Verizon: LTE Is Back and Euronews: April 21.)

- T-Mobile Deutschland GmbH , April 2009: The German giant's 40 million mobile customers were cut off for about four hours following a HLR system crash. The vendor supplier, NSN, held up its hands and issued an apology, a move not only to be applauded but one that would have helped T-Mobile get to grips with the fallout. (See Server Glitch Crashes T-Mobile Network.)

Signaling storm : EE 4G network - March,2014



Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices



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Since 4G deployment, we face complex outages because of always-on devices

Trigger

- A failure in the network
- Devices are detached

Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices

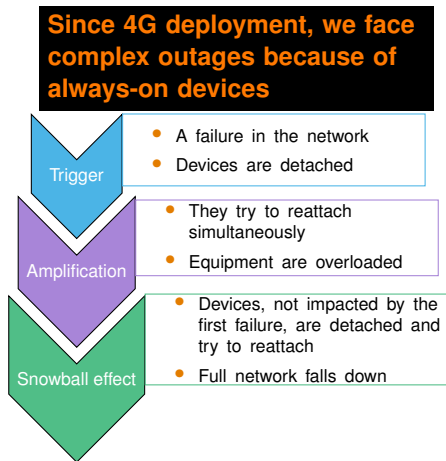
Trigger

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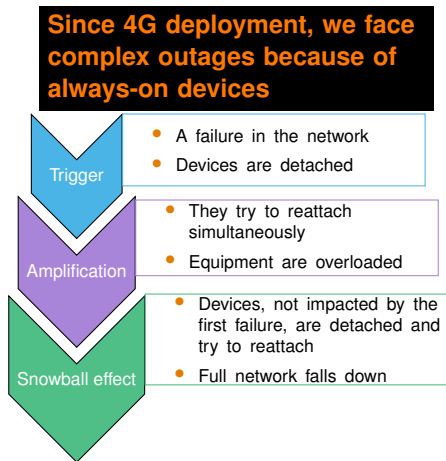
Amplification

- They try to reattach simultaneously
- Equipment are overloaded

Redundancy and diversity were not sufficient anymore in 4G



Redundancy and diversity were not sufficient anymore in 4G



4G standards improvement /
coordination mechanisms - timers

Industrial Implementation

Deployment

**Lessons learned for
5G standards design**

Resilience improvement for 5G

Redundancy/diversity

1

User Plan

New mechanisms allow a real user plan redundancy from an E2E perspective, combining access and core mechanisms

Redundancy/diversity

2

NF sets

NF (Network Function) can be organized as group of instances to allow load balancing, to react in case of failure of one instance

Load/Overload control

3

NF informs

Local or global mechanisms can be launched by an overloaded NF alerting its "neighbors" about its situation : timers, prioritization,...

4

Service continuity

Stateless NF

States information are stored in a separated database. In case of NF failure, another can take the floor retrieving the information from the database. No impact on the service

5

Dynamicity

Self healing...

Dynamicity is ensured by restoration mechanisms at 5G level and/or by NF instantiation on the fly at virtual machines or containers level

A 5G network is a system of systems

Network resilience can be affected by many parameters

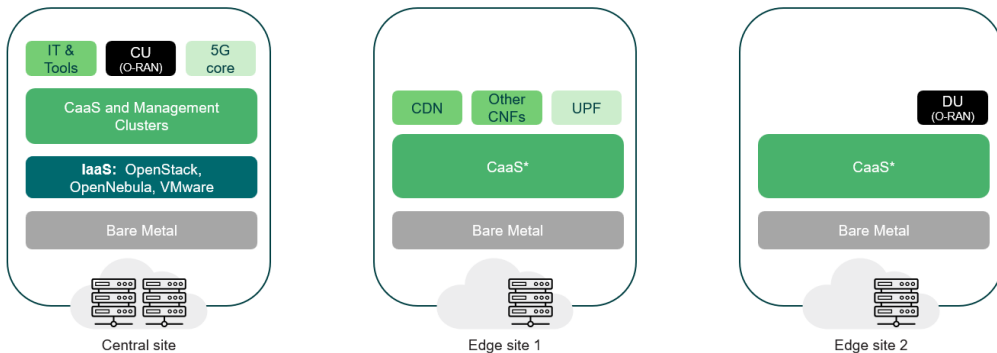
- hardware issues
- software issues (configuration, bug, ...)
- user traffic
- lifecycle management: updates, upgrades,
- weather conditions

5G and beyond networks are now built different

- Virtualization with Network Function Virtualization (NFV) and Software-Defined Networking (SDN)
- Slicing
- Cloud native microservices



New sources of problems join the old ones



from <https://sylvaproject.org/architecture-benefits/>



New sources of problems

New problems to address for resilience

- More layers
- More dependencies
- More dynamicity
- More heterogeneity



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but... Telco world is not fast fashion

- End of copper lines in France planned for 2030
- 2G and 3G, ... sunset while deploying newer generations
- Hardware, Software life cycles / Technical debt



New sources of problems

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- More heterogeneity

but... Telco world is not fast fashion

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Agence nationale de la sécurité
des systèmes d'information
ANSSI



Inspection générale de
l'administration
IGAD



Conseil général de l'économie
CGE



Commissariat aux communications
électroniques de défense



Inspection générale des
affaires sociales
IGAS

RAPPORT

ÉVALUATION DE LA GESTION PAR L'OPÉRATEUR ORANGE
DE LA PANNE DU 2 JUIN ET DE SES CONSÉQUENCES
SUR L'ACCÈS AUX SERVICES D'URGENCE

Paris, le 19 juillet 2021

51 boulevard de La Tour-Maubourg - 75006 PARIS 06 SP

5G Requirements (some examples)

Scenario	Max allowed end-to-end latency	Survival time	Communication Service availability	Reliability
Wireless road-size infrastructure backhaul	30 ms	100 ms	99.9999%	99.999 %
Distributed automated switching for isolation and service restoration of the power grid	5 ms	1 frame loss	99.9999%	
Motion control in factory	< transfer interval value	500 μ s to 2 ms	99.999% 99.9999%	to MTBF = 10 years
Control of automated train	< 100ms		99.999%	MTBF = 1 year
Very critical data communication for rail traffic mgmt	< 100ms			99.9999%

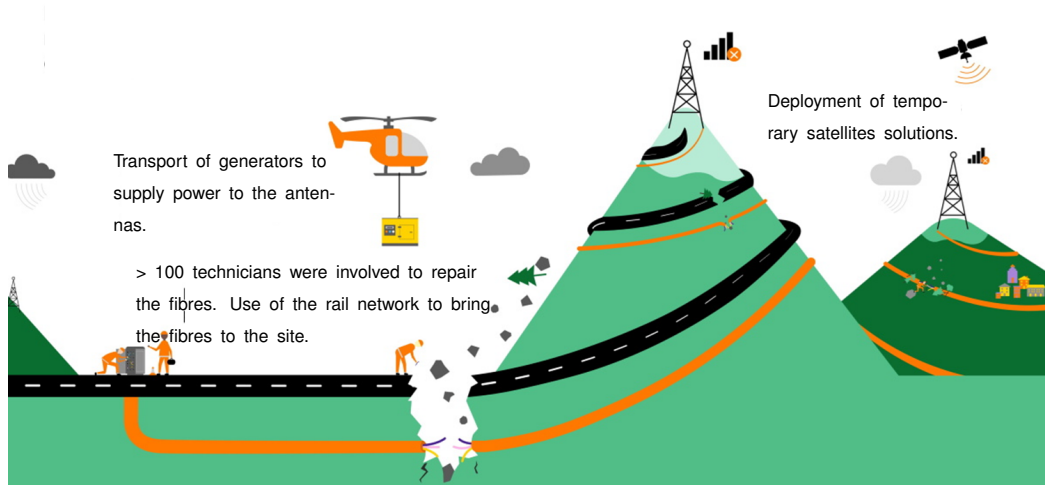
3GPP sources

- TS 22.261 - Service requirements for the 5G system
- TS 22.104 - Service requirements for cyber-physical control applications in vertical domains
- TS 22.289 - Mobile communication system for railways
- TS 22.186 - Service requirements for enhanced V2X scenarios
- TR 23.725 - Study on enhancement of Ultra-Reliable Low-Latency Communication (URLLC) support in the 5G Core network (5GC)

Flood in la vallée de la Vésubie, October 2020



Flood in la vallée de la Vésubie, October 2020



New, and future, problems



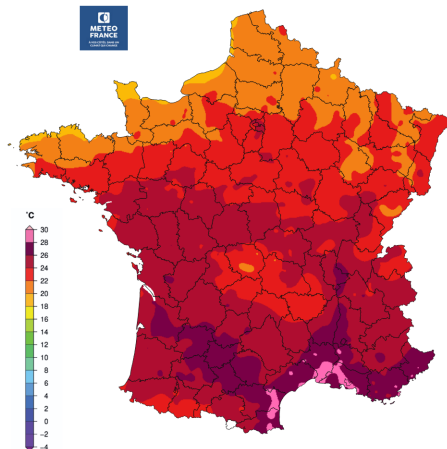
Mayotte, a French department in the Indian Ocean, was hit by tropical cyclone Chido on December 14th, 2024.

51 out of 54 mobile network antennas are out of service, mainly due to a lack of power. **99% of fixed-line Internet customers** are disconnected.

Increased complexity for the coordination of emergency services.

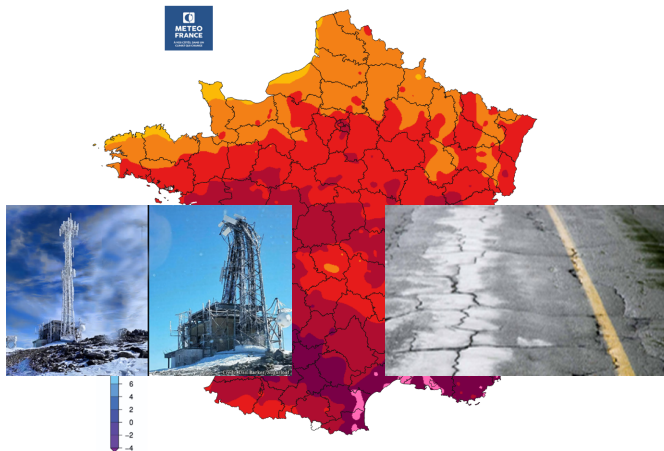


New, and future, problems



July 2026 is the hottest month ever recorded in France, average temperature was **24.9°C**. Climate change leads to higher average temperatures, longer heat waves, increased risks of fires, . . . Impact on material, infrastructure, people, ...

New, and future, problems



July 2026 is the hottest month ever recorded in France, average temperature was **24.9°C**. Climate change leads to higher temperatures, heat waves, Impact people,



Global warming is not an hoax



PLANÈTE • RÉCHAUFFEMENT CLIMATIQUE

Août 2026 est le mois le plus chaud jamais mesuré dans le monde, Copernicus alerte sur des risques accrus pour l'Europe



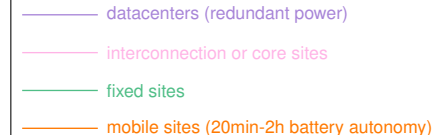
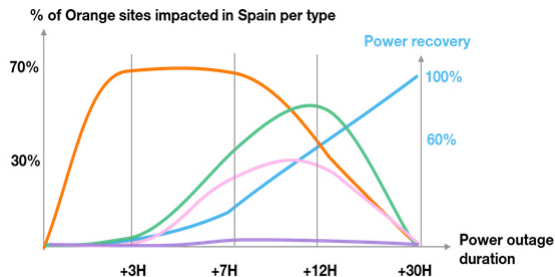
Orange's commitment: net zero carbon by 2040

- 49.3% reduction of CO² in Scopes 1 and 2 between in 2025 (vs. 2015)
- 54% electricity from renewable energy in 2025
- 14 000 solarized sites in MEA
- 80% of greenhouse gas emissions come from Scope 3

Exogenous causes and consequences

Networks are interdependent

- Telecommunication networks require energy / Energy networks require transmission of information
- Transportation needed to access sites / Vehicles and infrastructure require connectivity



Impact of April 2025 power blackout in Spain on Orange sites, *Revue de l'Electricité et de l'Electronique* 01/2026

Modeling Formalisms

Non state-space models

- Fast and simple to solve
- Limited ability to capture dynamic behavior

Examples: RBDs, reliability graphs, fault trees

State-space models

- Capture state, time, and component dependencies
- May suffer from **state-space explosion**

Examples: CTMCs, Petri nets

Multi-level models

- Combine state-space and non state-space models
- Suitable for large and complex infrastructures

Also called hierarchical models

Key trade-off

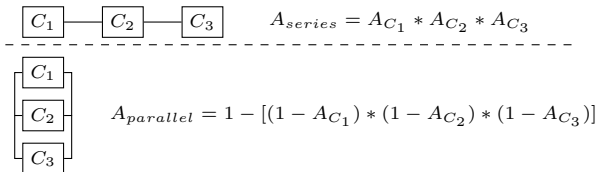
Simple models are efficient but less expressive; dynamic models are more realistic but can become difficult to analyze as the number of states grows.

Non State-Space Models: RBDs and Fault Trees

Reliability Block Diagrams (RBDs)

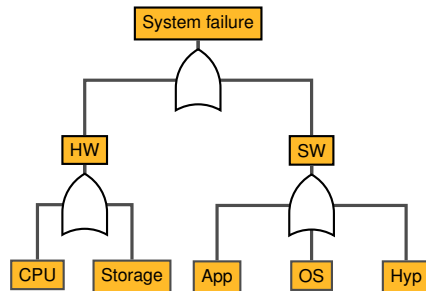
An RBD represents the logical arrangement of components required for successful system operation. Each component is either **operational** or **failed**.

- **Series:** all components must operate.
- **Parallel:** at least one component must operate.
- **k -out-of- n :** at least k components out of n must operate.



Fault Trees (FTs)

- Connects component failures through logic gates;
- **OR gates:** one event can cause failure;
- **AND gates:** multiple events are required.

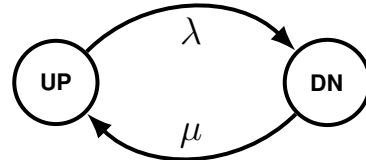


Continuous-Time Markov Chains (CTMCs)

CTMC principle

A CTMC models a system as a set of states and probabilistic transitions occurring at any time.

- **UP**: system is operational;
- **DN**: system is in a failed state;
- λ : failure rate, from UP to DN;
- μ : repair/recovery rate, from DN to UP.



Steady-state availability

For the two-state model:

$$\lambda\pi_{UP} = \mu\pi_{DN}, \quad \pi_{UP} + \pi_{DN} = 1$$

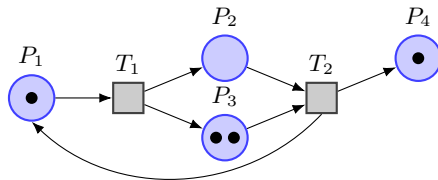
$$A_{sys} = \pi_{UP} = \frac{\mu}{\lambda + \mu} = \frac{MTTF}{MTTF + MTTR}$$

Petri Nets and Stochastic Reward Nets

Petri Nets (PNs)

- Directed bipartite graphs composed of **places** and **transitions**;
- Tokens** represent the current state of the system;
- Token movements capture dynamic system behavior.

Extensions: SPN, GSPN, SRN...



Stochastic Reward Nets (SRNs)

SRNs extend availability analysis by associating a reward with each system state (marking). several measures can be calculated using the reward function (e.g. For availability, $r(t)=1$ in operational states and 0 otherwise).

Steady-state reward measure

$$E[X] = \sum_{j \in \mathcal{S}} r_j \cdot \pi_j$$

r_j : reward of marking j

π_j : steady-state probability of marking j

State of the Art (partial)

Availability modeling in 5G networks

Reference(s)	Model	Network domain / service
Kaada et al., 2022, 2023 [2, 3]	CTMC	RAN
Temericht et al., 2025 [4]	RBD + CTMC	5G Service Function Chain
Soares et al., 2019[5]	RBD–SAN	Virtualized services / VNFs
Matos et al., 2019 [6]	RBD–FT–CTMC	IP Multimedia Subsystem
Khorsandroo et al., 2021 [7]	RBD–SRN	PDU session establishment
Dantas et al., 2021 [8]	FT–CTMC	Medical edge network
Matos et al., 2017 [9]	RG–FT–SRN	Cloud data-center network
Pathirana et al., 2024 [10]	FT–SAN	5G-MEC / O-RAN
Li et al., 2022, 2023 [11, 12]	Three-level TSCPN	SFC, VNF, and microservice layers

Main observations

- Strong focus on **SFCs**, VNFs, microservices, cloud, edge, and 5G Core services.
- Various dependability models are used: **CTMC**, **RBD**, **FT**, **SAN**, **SRN**, and Petri nets.
- Limited number of work on **RAN**.

Aim

- Find exact or good estimates availability of the system (or service) availability (analytical expressions a plus)
- Identify the key parameters of the problem under consideration
- Go beyond the Markov approximation, which is not always realistic and may overestimate reliability
- Provide a better understanding that can be applied in ensuing simulations and policies (digital twins, maintenance)



Selected research activities



Meshed networks [14]

The exact expression of the end-to-end availability between S and T can be obtained, not by using series-parallel simplifications, but from a generic solution for recursive families of graphs (917 terms)

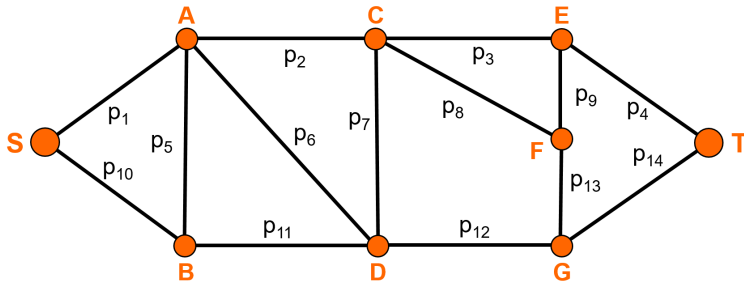


figure from [13]



Meshed networks [14]

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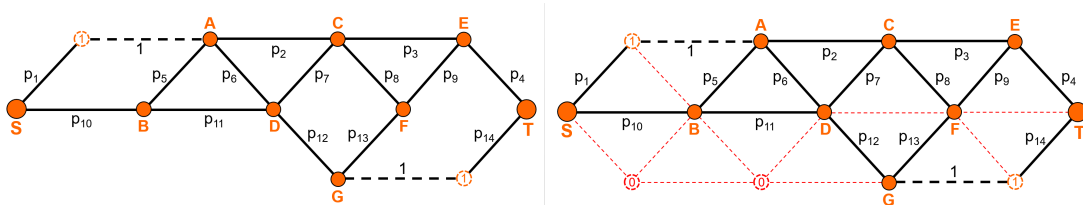
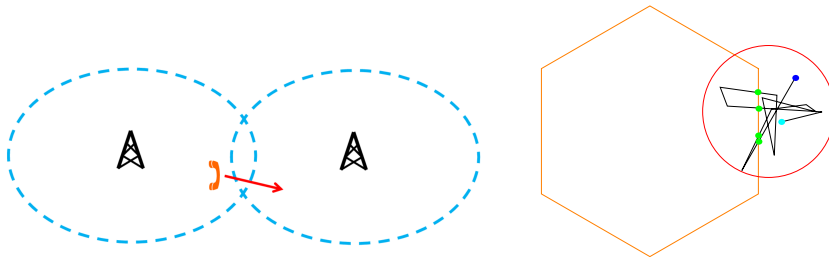


figure from [13]



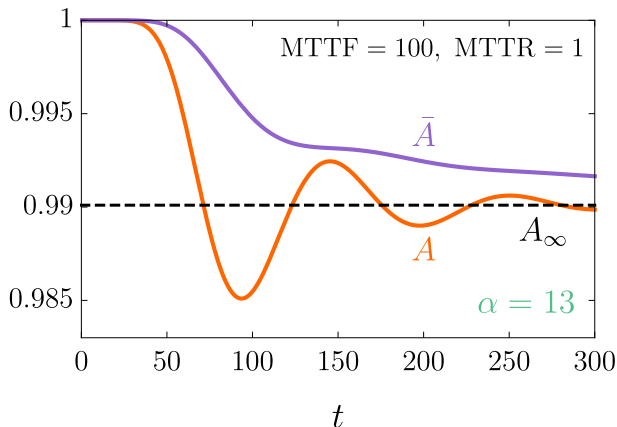
Hand-over rate and Random Waypoint model [15]



- What is the probability of leaving a cell?

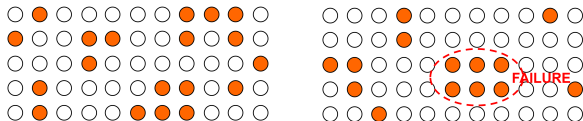


Importance of transients [16]

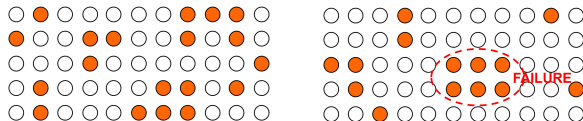


Instant and averaged availabilities $A(t)$ and $\bar{A}(t)$ for a gamma($\lambda, 13$) lifetime distribution and an exponential repair distribution

Two-dimensional (3,2)-out-of-(11,5):F system [17]



Two-dimensional (3,2)-out-of-(11,5):F system [17]



$$\begin{aligned}
 \mathcal{R} = & (1 - q)^6 \left(1 + 6q + 21q^2 + 56q^3 + 126q^4 + 252q^5 + 426q^6 + 576q^7 \right. \\
 & + 563q^8 + 205q^9 - 651q^{10} - 2111q^{11} - 3865q^{12} - 4912q^{13} - 3949q^{14} \\
 & - 111q^{15} + 5867q^{16} + 12046q^{17} + 15332q^{18} + 12047q^{19} + 1086q^{20} \\
 & - 14387q^{21} - 25509q^{22} - 24554q^{23} - 11602q^{24} + 7156q^{25} + 22184q^{26} \\
 & + 27253q^{27} + 18578q^{28} - 240q^{29} - 16433q^{30} - 19156q^{31} - 9994q^{32} \\
 & + 856q^{33} + 6078q^{34} + 5867q^{35} + 3723q^{36} + 731q^{37} - 1794q^{38} \\
 & - 1925q^{39} - 344q^{40} + 406q^{41} + 69q^{42} - 38q^{43} + 21q^{44} \\
 & \left. + 41q^{45} + 12q^{46} - 4q^{47} - 4q^{48} - q^{49} \right)
 \end{aligned}$$

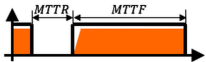
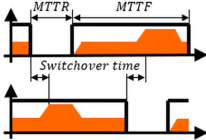
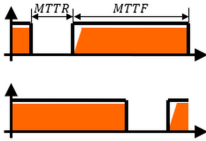
where q is the unavailability of each element



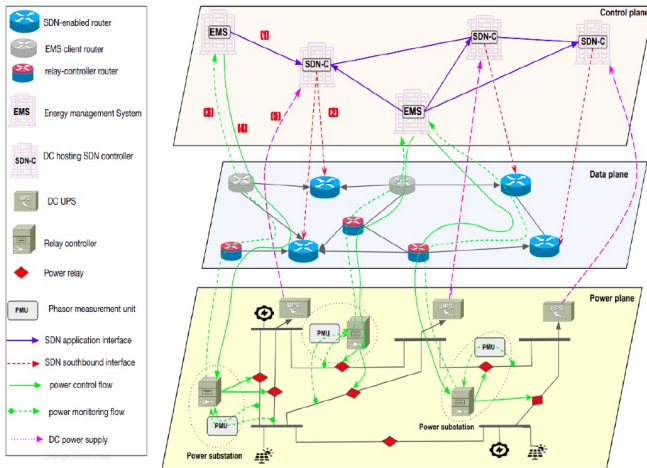
Selected publications

- [14] Christian Tanguy. “What is the probability of connecting two points?” In: *Journal of Physics A: Mathematical and Theoretical* 40.47 (2007), pp. 14099–14116.
- [15] Christian Tanguy. “Handover Rate in Wireless Networks and Random Waypoint Mobility Model: An Almost Simple Result”. In: 10th International Conference on Mathematical Methods in Reliability (Grenoble, France). 2017.
- [16] Christian Tanguy. “Transient Behaviour of Instantaneous and Average Availabilities in Non-Markovian Configuration”. In: *Safety and Reliability of Systems and Processes* (2022), pp. 185–194. DOI: [10.26408/srsp-2022-13](https://doi.org/10.26408/srsp-2022-13).
- [17] Christian Tanguy. “Impact of Common-Cause Failures on the Availability of Connected (r, s)-out-of-(m, n): F Systems”. In: *35th European Safety and Reliability Conference (ESREL 2025) and the 33rd Society for Risk Analysis Europe Conference (SRA-E 2025)*. 35th European Safety and Reliability Conference (ESREL 2025) (Stavanger, Norway). RESEARCH PUBLISHING, SINGAPORE, 2025. ISBN: 978-981-94-3281-3. DOI: [10.3850/978-981-94-3281-3-proceedings](https://doi.org/10.3850/978-981-94-3281-3-proceedings)

Typical resilience mechanisms for telcos

		Scheme	Backup ratio	Failure-Repair Sequence	Interruption Time
	App. Server	Self Healing	0		MTTR
	5G User Plane	N:B Shared Protection	B/N		switchover time
 	5G Control Plane	1+1 active/active protection	1		0

Cross-domain resilience in Cloud-native, Critical Cyber Physical Systems Networks [28]



Results: Model for the propagation of failures between **interdependent** critical infrastructures and analyze its impact on overall availability. Optimize the placement of virtualized functions across multiple sites based on availability and latency constraints. Propose an architecture for exchanging availability information to plan maintenance operations.

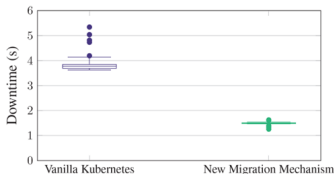
Supervised Learning of virtualization infrastructure availability

Objective: Model the failure statistics of infrastructure components and develop a supervised learning approach for the model parameters-aimed at compressing measurements from infrastructure sensors for a Digital Twin-while retaining only information regarding rare events such as failures, resilience, and the return to normal operations.

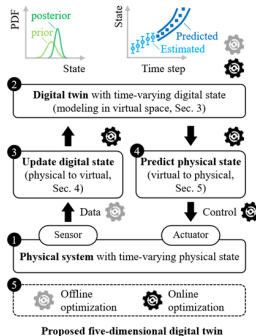
Tier	a_d	Monthly DC downtime
Tier 4	0.9999	4 min 19 s
Tier 3	0.9998	8 min 38 s
Tier 2	0.9975	1 h 48 min
Tier 1	0.9967	42 h 23 min



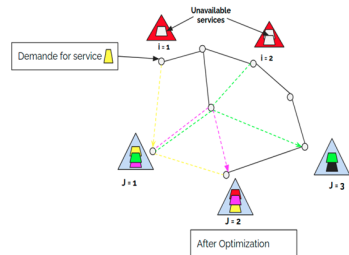
from [38]



from [36]



from [37]



from [29]

Selected publications

- [28] Khaled Sayad. “Cross-Domain Resilience in Cloud-native, Critical Cyber-Physical Systems Networks : Availability Modeling, Analysis, and Optimization of Critical Services Provisioning”. These de doctorat. université Paris-Saclay, Apr. 10, 2024. URL: <https://theses.fr/2024UPAST028>.
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From availability assessment to automation

Availability can be evaluated using several **well-established approaches**.



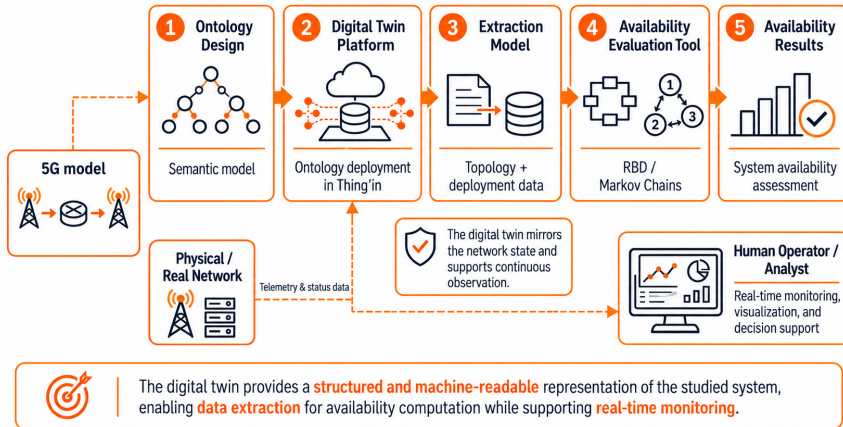
Practical challenge

The methods themselves are mature, but building a complete end-to-end assessment often requires manual effort for **model construction • scenario configuration • failure injection • data collection • analysis**.



Digital Twins provide a promising foundation for making availability assessment **systematic, repeatable and automatable**.

Digital twin technology



Ontology Design for the 5G Digital Twin

The ontology provides a **semantic representation** of the main entities, relationships and attributes of the virtualized 5G system.

Ontology classes

VNF ● Microservice ● Replica ● Pod ● Server

Object properties

hasMicroservice: VNF → Microservice

hasReplica: Microservice → Replica

deployedIn: Replica → Pod

hostedOn: Pod → Server

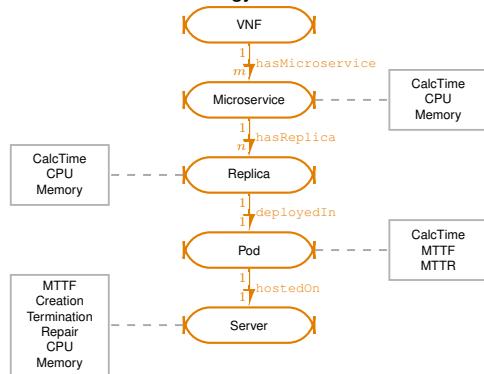
Data properties

Microservice / Replica: CalcTime, CPU, Memory

Pod: MTTF, MTTR, CalcTime

Server: MTTF, MTTR, Creation, Termination, CPU, Memory

Ontology schema



The ontology captures both the **logical service structure** and its **deployment infrastructure**.

From Digital Twin Data to Availability Evaluation

Extracted topology & redundancy

```
VNF: CU
  MicroserviceCU1: 2 replicas
  MicroserviceCU2: 2 replicas
  MicroserviceCU3: 2 replicas
VNF: DU
  MicroserviceDU1: 2 replicas
  MicroserviceDU2: 2 replicas
  MicroserviceDU3: 2 replicas
VNF: UPF
  Microservice1: 2 replicas
  Microservice2: 2 replicas
  Microservice3: 2 replicas
```

Extracted dependability parameters

Availability input parameters

```
Server MTTF      : 5000
Server repairTime: 1
Pod MTTF         : 200
Pod repairTime   : 0.00278
```

Failure and repair rates

```
lambda_server = 2.000000e-04 /h
mu_server     = 1.000000e+00 /h
lambda_pod    = 5.000000e-03 /h
mu_pod        = 3.597122e+02 /h
```

Inputs to the availability model

Availability Evaluation
RBD / Markov Chains

System Availability

99.999917%

Use case: TMForum : Living Network – Phase III Catalyst



The studied use case is composed of two service chains: a **User Plane Chain** and a **Control Plane Chain**.

User Plane Chain



Control Plane Chain



Modeling assumptions:

each **NF/VNF** is decomposed into **3 microservices**,
and each microservice is instantiated with **2 replicas**.

Selected publications

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Other actions attached to a chair

Risk and Resilience of Complex Systems (RRSC) Chair of CentraleSupélec, led by Prof. Anne Barros https://r3.centralesupelec.fr/en/Chair_RRCS



Scientific Project 2024-2029:

- Axis 1: System Modeling for Risk, Resilience, and Reliability
- Axis 2: Decision Support under Uncertainty
- Axis 3: Environmental and Societal Challenges



Partners



naTran



谢谢

Thank you

Merci



*"The greatest glory
of living lies not
in never falling but
in rising every time
you fall."*

Nelson Mandela



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